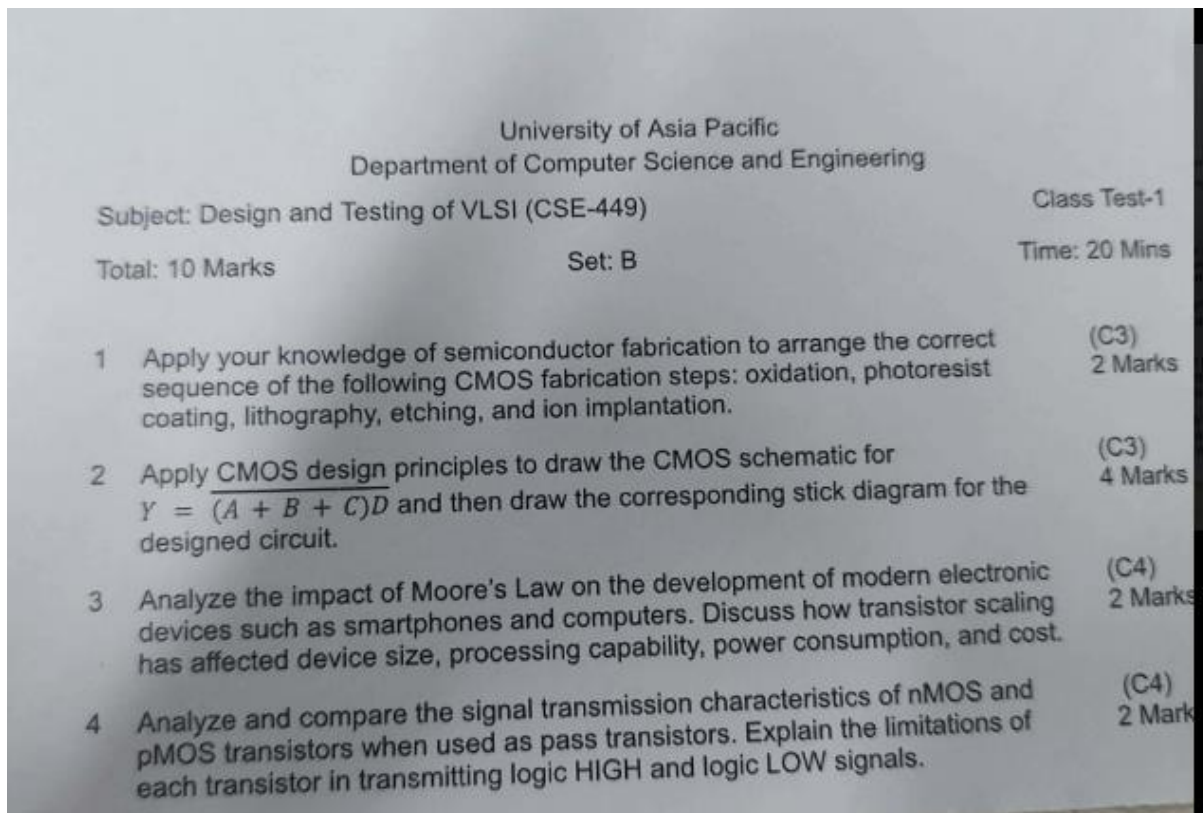


VLSI MID 13 September 2026



VLSI Class Test-1 — Set B Solution

1. CMOS Fabrication Step Sequence — 2 Marks

The correct sequence is:

1. Oxidation → 2. Photoresist coating → 3. Lithography → 4. Etching → 5. Ion implantation

Explanation:

1. **Oxidation:** A thin layer of SiO_2 is grown on the silicon wafer.
2. **Photoresist coating:** Photoresist is applied over the oxide layer.
3. **Lithography:** UV light and a mask are used to transfer the required pattern to the photoresist.

4. **Etching:** The exposed oxide/material is removed according to the lithographic pattern.
5. **Ion implantation:** Dopant ions such as boron or phosphorus are implanted into selected regions to form **p-type/n-type regions**.

Answer:

Oxidation → Photoresist → Lithography → Etching → Ion Implantation

2. CMOS Schematic and Stick Diagram — 4 Marks

Given:

$$Y = \overline{(A+B+C)D}$$

We first determine the condition when the output becomes **0**:

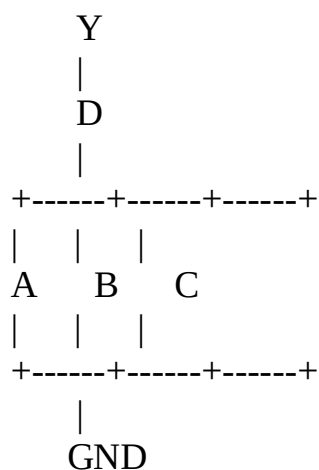
$$Y=0 \text{ when } (A+B+C)D=1 \quad Y=0 \quad \text{when} \quad (A+B+C)D=1$$

Therefore, the **NMOS Pull-Down Network (PDN)** must implement:

$$D(A+B+C)$$

NMOS Pull-Down Network

- DD is **in series** with
- A,B,C in **parallel**



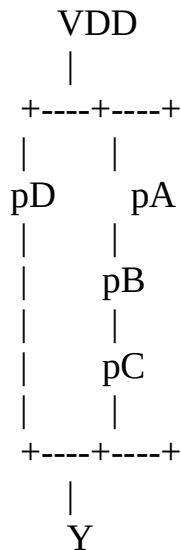
So:

$PDN = D \text{ series } (A \parallel B \parallel C)$

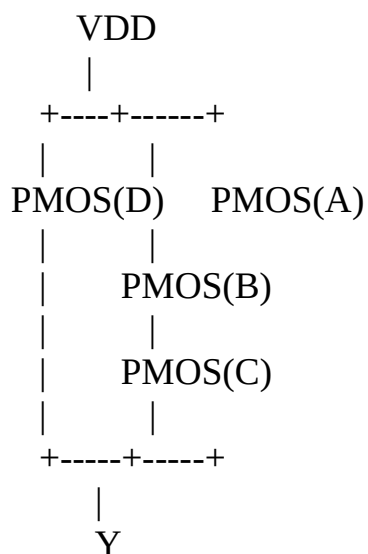
PMOS Pull-Up Network

The PMOS network is the **dual** of the NMOS network:

- DD is in **parallel** with
- A,B,C in **series**



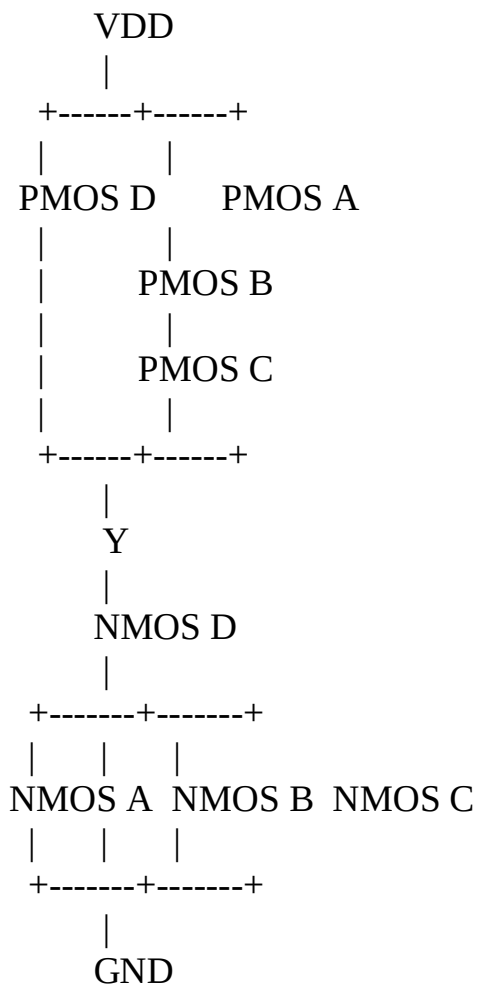
More clearly:



Thus,

$PUN = D \parallel (A - B - C)$

Final CMOS structure

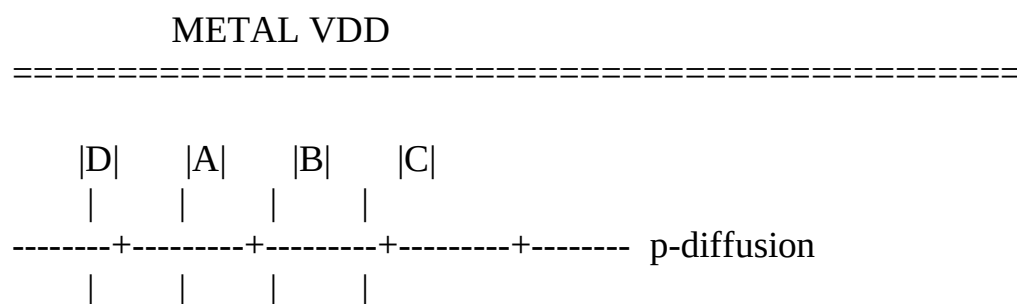


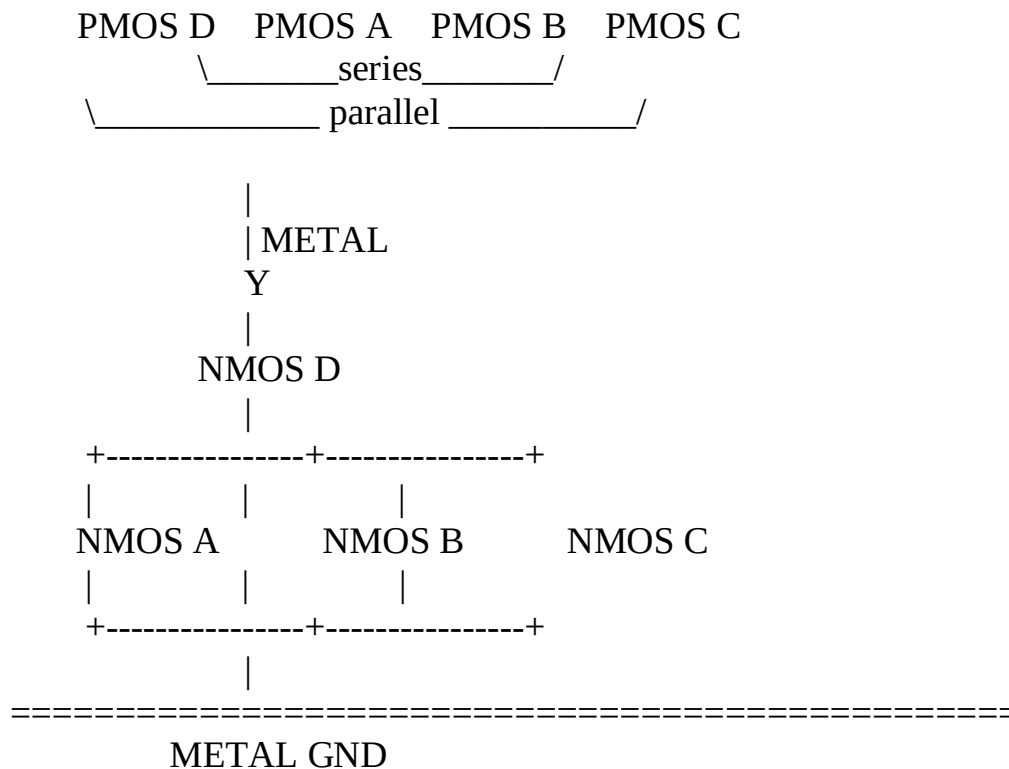
Stick Diagram

Using the usual stick-diagram convention:

- **Metal:** horizontal/vertical line
- **Polysilicon:** vertical lines crossing diffusion
- **n-diffusion:** NMOS region
- **p-diffusion:** PMOS region

A simplified stick representation is:





Important for the exam: Remember the key rule:

NMOS: series $\leftrightarrow$ AND, parallel $\leftrightarrow$ OR

PMOS: use the **dual** network.

3. Impact of Moore's Law — 2 Marks

Moore's Law states that the number of transistors on an integrated circuit tends to increase exponentially over time, historically roughly doubling every two years.

Effects:

- **Smaller devices:** Transistor scaling allows smartphones and computers to become smaller and more compact.
- **Higher processing capability:** More transistors provide more CPU/GPU cores, larger caches, and greater computational power.
- **Lower power per operation:** Smaller and more efficient transistors can reduce energy consumption for individual operations, although total chip power can still be high because of increased transistor count.
- **Lower cost per function:** More transistors can be integrated onto one chip, reducing the cost per computational function and enabling inexpensive, powerful electronics.

Conclusion: Moore's Law has driven the development of **smaller, faster, more capable, and increasingly affordable** electronic devices.

4. nMOS vs pMOS as Pass Transistors — 2 Marks

Transistor Passes Logic 0 Passes Logic 1

nMOS	Strong 0	Weak 1
pMOS	Weak 0	Strong 1

nMOS Pass Transistor

An nMOS conducts well when its gate is HIGH.

- Logic **0** passes strongly → output can reach approximately 0V0V.
- Logic **1** is passed weakly because the output can rise only to approximately:

$$V_{DD} - V_{TNV_DD} - V_{TN}$$

So nMOS cannot transmit a full logic HIGH.

pMOS Pass Transistor

A pMOS conducts well when its gate is LOW.

- Logic **1** passes strongly → output can reach approximately V_{DDV_DD} .
- Logic **0** is passed weakly because the output can fall only to approximately:

$$|V_{TP}| |V_{TP}|$$

above ground.

Final point

nMOS = strong 0, weak 1
pMOS = strong 1, weak 0

Therefore, **transmission gates**, which combine an nMOS and pMOS, are commonly used to pass both logic HIGH and LOW strongly.

Department of Computer Science and Engineering		
Subject: Design and Testing of VLSI (CSE-449)		Class Test-2
Total: 10 Marks	Set: B	Time: 25 Mins
1	Differentiate the accumulation and depletion modes of an NMOS transistor based on the applied gate-to-source voltage and channel formation.	(C4) 2 Marks
2	Investigate the effects of substrate doping and SiO ₂ impurities on the threshold voltage V_{TH} of a MOSFET.	(C4) 2 Marks
3	Consider an nMOS transistor in a 65 nm process working with the given configuration. $V_{gs} = 0.6$, $V_{ds} = 0.8$, $V_t = 0.3$. Given process-transistor parameter, $\beta = 1.31 \times 10^{-5} \text{ A/V}^2$. Identify the operation region and calculate the current I_{ds} .	(C3) 3 Marks
4	Consider the nMOS transistor in a 65 nm process with a nominal threshold voltage [When $V_{sb} = 0$] of 0.3 V and a doping level of $8 \times 10^{17} \text{ cm}^{-3}$. The body is tied to ground with a substrate contact. Calculate the change in threshold voltage at room temperature if the source is at 0.9 V instead of 0. [Given intrinsic carrier concentration $N_i = 1.55 \times 10^{10} \text{ cm}^{-3}$, body effect coefficient $\gamma = 0.68$, equation for calculating threshold voltage $V_t = V_{t0} + \gamma(\sqrt{2\phi_s + V_{sb}} - \sqrt{2\phi_s})$	(C3) 3 Marks

VLSI Class Test-2 — Set B Solution

1. Accumulation vs. Depletion Mode of an NMOS — 2 Marks

For an **NMOS**, the substrate is p-type.

Accumulation mode

- When $V_{GS} < 0$, negative gate voltage attracts **holes** toward the Si-SiO₂ interface.
- The majority carriers (holes) accumulate near the surface.
- **No n-type inversion channel** is formed.

Depletion mode

- When $V_{GS} > 0$ but $V_{GS} < V_{T0}$, positive gate voltage repels holes from the surface.
- A **depletion region** containing fixed ionized acceptor ions is formed.
- Since $V_{GS} < V_{T0}$, a conducting inversion channel has not yet formed.

When:

$$V_{GS} \geq V_{T0} \Rightarrow V_{GS} \geq V_T$$

an **n-type inversion channel** forms between source and drain, and the NMOS conducts.

Short answer

Mode	V_{GS}	Surface condition
Accumulation	$V_{GS} < 0$	Holes accumulate
Depletion	$0 < V_{GS} < V_{T0}$	Holes are repelled; depletion region forms
Inversion	$V_{GS} > V_T$	n-type channel forms

2. Effects of Substrate Doping and SiO₂ Impurities on V_T — 2 Marks

The threshold voltage of an NMOS depends strongly on substrate doping and oxide charge.

Effect of substrate doping

For a p-type substrate:

$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{4q\epsilon_{Si}N_A\phi_F}}{C_{ox}} \Rightarrow V_T = V_{FB} + 2\phi_F + \frac{\sqrt{4q\epsilon_{Si}N_A\phi_F}}{C_{ox}}$$

where

$$\phi_F = kT \ln\left(\frac{N_A}{n_i}\right) \Rightarrow \phi_F = \frac{kT}{q} \ln\left(\frac{N_A}{n_i}\right)$$

As the substrate doping N_A **increases**, the magnitude of the threshold voltage generally **increases**.

So:

$$N_A \uparrow \Rightarrow V_T \uparrow \quad \boxed{N_A \uparrow \Rightarrow V_T \uparrow}$$

Effect of SiO₂ impurities

Charges/impurities inside SiO₂ or at the Si–SiO₂ interface modify the flat-band voltage:

$$V_{FB} = \phi_{MS} - Q_{ox} C_{ox} V_{FB} = \phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

Therefore, oxide/interface charge can **shift** V_T .

For example, positive oxide charge $Q_{ox} > 0$ generally causes:

$$V_T \downarrow \boxed{V_T \downarrow}$$

for an NMOS.

3. NMOS Operating Region and I_{DS} — 3 Marks

Given:

$$V_{GS} = 0.6V \quad V_{DS} = 0.8V \quad V_T = 0.3V \\ \beta = 1.31 \times 10^{-5} \text{ A/V}^2$$

Step 1: Calculate overdrive voltage

$$V_{OV} = V_{GS} - V_T = 0.6V - 0.3V = 0.3V \\ \boxed{V_{OV} = 0.3V}$$

Step 2: Determine operating region

For an NMOS:

- Linear/triode region:

$$V_{DS} < V_{GS} - V_T$$

- Saturation region:

$$V_{DS} \geq V_{GS} - V_T$$

Here:

$$0.8 > 0.3$$

Therefore:

NMOS is operating in saturation region\boxed{\text{NMOS is operating in saturation region}}

Step 3: Calculate drain current

In saturation:

$$I_{DS} = \frac{\beta}{2} (V_{GS} - V_T)^2$$

Substitute:

$$I_{DS} = \frac{1.31 \times 10^{-5}}{2} (0.6 - 0.3)^2$$

$$I_{DS} = 6.55 \times 10^{-6} \times 0.09$$

$$I_{DS} = 5.895 \times 10^{-7} \text{ A}$$

Therefore:

$$I_{DS} \approx 5.90 \times 10^{-7} \text{ A} \boxed{I_{DS} \approx 5.90 \times 10^{-7} \text{ A}}$$

or

$$I_{DS} \approx 0.590 \text{ } \mu\text{A} \boxed{I_{DS} \approx 0.590 \text{ } \mu\text{A}}$$

Final answer:

Region = Saturation\boxed{\text{Region = Saturation}} $I_{DS} \approx 0.59 \text{ } \mu\text{A} \boxed{I_{DS} \approx 0.59 \text{ } \mu\text{A}}$

4. Body Effect and Change in Threshold Voltage — 3 Marks

Given:

$$\begin{aligned} V_{T0} &= 0.3 \text{ V} & V_{T0} &= 0.3 \text{ V} & N_A &= 8 \times 10^{17} \text{ cm}^{-3} & N_A &= 8 \times 10^{17} \text{ cm}^{-3} \\ V_{SB} &= 0.9 \text{ V} & V_{SB} &= 0.9 \text{ V} & n_i &= 1.55 \times 10^{10} \text{ cm}^{-3} & n_i &= 1.55 \times 10^{10} \text{ cm}^{-3} \\ \gamma &= 0.68 & \gamma &= 0.68 \end{aligned}$$

At room temperature:

$$V_{\text{thermal}} = kT/q \approx 0.0259 \text{ V} \quad V_{\text{thermal}} = \frac{kT}{q} \approx 0.0259 \text{ V}$$

The given equation is:

$$V_T = V_{T0} + \gamma [2\phi_s + V_{SB} - 2\phi_s] \quad V_T = V_{T0} + \gamma \left[\sqrt{2\phi_s + V_{SB}} - \sqrt{2\phi_s} \right]$$

We first calculate ϕ_s .

Step 1: Calculate Fermi potential

For a p-type substrate:

$$\phi_s = 2kTq \ln \left(\frac{N_A}{n_i} \right) \quad \phi_s = 2 \frac{kT}{q} \ln \left(\frac{N_A}{n_i} \right)$$

Therefore:

$$\begin{aligned} \phi_s &= 2(0.0259) \ln \left(\frac{8 \times 10^{17}}{1.55 \times 10^{10}} \right) \quad \phi_s = 2(0.0259) \\ &\ln \left(\frac{8 \times 10^{17}}{1.55 \times 10^{10}} \right) \\ 8 \times 10^{17} / 1.55 \times 10^{10} &\approx 5.161 \times 10^7 \quad \frac{8 \times 10^{17}}{1.55 \times 10^{10}} \\ &\approx 5.161 \times 10^7 \end{aligned}$$

Thus:

$$\phi_s \approx 0.920V \quad \phi_s \approx 0.920V$$

So:

$$\phi_s \approx 0.92V \quad \boxed{\phi_s \approx 0.92V}$$

Hence:

$$2\phi_s \approx 1.84V \quad 2\phi_s \approx 1.84V$$

Step 2: Calculate threshold voltage change

$$\Delta V_T = \gamma [2\phi_s + V_{SB} - 2\phi_s] \quad \Delta V_T = \gamma \left[\sqrt{2\phi_s + V_{SB}} - \sqrt{2\phi_s} \right]$$

Substitute:

$$\begin{aligned} \Delta V_T &= 0.68 [1.84 + 0.9 - 1.84] \quad \Delta V_T = 0.68 \left[\sqrt{1.84 + 0.9} - \sqrt{1.84} \right] \\ &= 0.68 [2.74 - 1.84] = 0.68 [\sqrt{2.74} - \sqrt{1.84}] \\ &= 0.68 [1.655 - 1.356] = 0.68 [1.655 - 1.356] \quad \Delta V_T \approx 0.203V \quad \Delta V_T \approx 0.203V \end{aligned}$$

So approximately:

$$\Delta V_T \approx 0.20V \quad \boxed{\Delta V_T \approx 0.20V}$$

Step 3: Calculate new threshold voltage

$$V_T = V_{T0} + \Delta V_T \quad V_T = 0.3 + 0.203 \quad V_T \approx 0.503V \quad \boxed{V_T \approx 0.503V}$$

Final answer:

$$\Delta V_T \approx 0.20V \quad \boxed{\Delta V_T \approx 0.20V}$$

$$V_T \approx 0.50V \quad \boxed{V_T \approx 0.50V}$$

Note: If your instructor's notation uses ϕ_s to mean the **surface potential** $2\phi_F$, the calculation is exactly the form above. Some textbooks instead define ϕ_F differently, so be careful with the factor of 2.

★ Quick Exam Revision

Question	Key Answer
1	Accumulation: $V_{GS} < 0$, holes accumulate. Depletion: $0 < V_{GS} < V_{T0}$, holes are repelled.
2	Higher substrate doping $\rightarrow$ generally higher V_T ; oxide/interface charge shifts V_T .
3	$V_{GS} - V_T = 0.3V$, $V_{DS} = 0.8V > 0.3V \rightarrow$ Saturation , $I_D \approx 0.59 \mu A$
4	$\phi_s \approx 0.92V$, $\Delta V_T \approx 0.20V$, new $V_T \approx 0.50V$

Monday 10:32 AM

1. Accumulation vs. Depletion Mode of an PMOS — 2 Marks

1. Accumulation vs. Depletion Mode of a PMOS — 2 Marks

For a **PMOS**, the body/substrate is **n-type**.

Accumulation Mode

- When the gate voltage is **positive relative to the source/body** ($V_{GS} > 0V_{GS} > 0$), electrons (majority carriers) are attracted toward the surface.
- This causes **accumulation of electrons** near the Si–SiO₂ interface.
- No p-type conducting channel is formed.

Depletion Mode

- When the gate voltage becomes **negative** relative to the source, electrons are repelled from the surface.
- A **depletion region** is formed.
- As the negative gate voltage becomes sufficiently large, holes are attracted to the surface and an **inversion layer (p-channel)** forms.

For PMOS, conduction occurs when:

$$V_{GS} < V_T \quad \boxed{V_{GS} < V_T}$$

where V_T is normally **negative**.

Short exam table

Mode	Gate voltage	Surface condition
Accumulation	$V_{GS} > 0V_{GS} > 0$	Electrons accumulate
Depletion	$V_T < V_{GS} < 0V_T < V_{GS} < 0$	Electrons are repelled; depletion region forms
Inversion	$V_{GS} < V_TV_{GS} < V_T$	Holes form a p-channel

Easy way to remember

NMOS:

Negative gate $\rightarrow$ Accumulation $\boxed{\text{Negative gate} \rightarrow \text{Accumulation}}$

PMOS:

Positive gate $\rightarrow$ Accumulation